

Repository URL: <https://github.com/Draconeez/GroupProject1>

Team Member	Component(s)	Specific Task Completed	Testing/Debugging	Github Contributions
Will	- Reservation.h - Reservation.cpp - main.cpp	Reservation Linked List Functionality Cleaned up inputs Linked list traversal Linear Search	Tested the functionality of the linked list in the reservation Tested for logic errors in Reservation Manager Fixed Milestone 1 bugs like stale availability and incorrect date presentation Tested Functionality in weekly milestones	- Added txt data - Implemented Reservation class and functions - Cleaned up readability in outputs - Debugged for logic errors - Part1Edits branch
KC	- CancellationHistory.h - CancellationHistory.cpp - ComplexityAnalysis.md	Wrote the stack logic for the undo feature and wrote the Big-O report.	Checked compatibility with Will's Reservation class to make sure empty stacks don't crash the program. Tested linux compatibility	- Uploaded CancellationHistory.h - CancellationHistory.cpp - ComplexityAnalysis.md
Aryn	- Resource.h - Resource.cpp - ReservationManager.h - ReservationManager.cpp - ReadME.md	Resource Management feature, waitlist functionality Insertion and Removal Menu traversal Developed quicksort Linear search Binary search And Reporting function Lots of comments	Tested functionality of producing and displaying resource status Tested menu traversal, and if each function worked Fixed Milestone 1 Bugs like waitlist promotion Undo duplication Testing functionality in weekly milestones	- Setup Files - Implemented classes - Directed forks for everyone to work on - Created ReportingFeatures Branch - Part2Features branch which included the searching and sorting algorithms

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Detailed Contributions:

Aryn - Resource.h, Resource.cpp, ReservationManager.h, ReservationManager.cpp - established the Resource class and set up the public constructor with id, name, type, and availability. ID, Name, and Type have their respective getters and setters. Established private resourceID, resourceName, resourceType, and availability. Additionally, created the ReservationManager class with the waitlist queue, where the cancellationhistory stack was implemented. In the class, added display, insertion, and removal from each of the three data structures. Finally, I created the main menu loading of the resources and reservations into the actual linked list and vector array, as well as using switch case to make a main menu where you can select each option.

(For Final)
Created linear search logic for linked lists and binary search for Resources, as well as quicksort algorithm for sorting requirement. Developed entire reporting feature, as well as added a bunch of comments explaining things throughout the files.

Will - Reservation.h, Reservation.cpp, main.cpp - established the Reservation class and set up the public constructor with ReservationID, studentID, name, ResourceID, and Date. They all have their respective getters and setters. Edited grammar and code to ensure quality and readability of the work. Implemented linked list traversal to make each node work through creation and destruction.

(For Final)
Fixed notable bugs shown by the TA, like incorrect presentation and stale availability, and standardized output.

KC - CancellationHistory.h & .cpp - established the CancellationHistory class for the undo feature. Set up the stack operations (push, pop, isEmpty, displayHistory) using a vector so it doesn't mess with the linked list pointers. ComplexityAnalysis.md - wrote the Big-O time complexity report for the project. Verified terminal compilation and full menu execution on Linux to ensure cross-platform stability. ComplexityAnalysis.md - wrote the Big-O time complexity report for the project.

(For Final)
Tested Code on linux.